

Emotion-Aware Bargaining System for Negotiation Training Based on Symmetrical Reality

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Figure 1: Illustration of the symmetrical reality bargaining framework, where a human in the physical world and an avatar in the virtual world engage in mirrored, bidirectional interaction. Through conversation, emotional expression, and mutual observation, both sides can perceive and influence each other's behavior in a shared negotiation space. This symmetrical coupling enables users to practice and refine their bargaining and communication skills in a controlled setting, while targeting improved performance in real-world negotiation scenarios.

ABSTRACT

Negotiation is a key skill in everyday consumption and workplace communication. However, most existing negotiation training tools rely on text based interfaces and rule driven scripts, providing limited situational context and insufficient support for practicing complex negotiation processes. To explore more context aware training approaches, we develop a Unity based virtual night market negotiation platform. The system consists of three core modules. A user preference modeling module estimates pricing preferences and concession tendencies from user inputs, historical bidding records, and interaction level cues, including linguistic style features and facial expression related visual indicators derived from image processing. An environmental factor modeling module represents predefined scene level variables such as noise level, crowd density, lighting conditions, visual distractions, and time constraints as contextual factors relevant to negotiation. A strategy selection and language generation module integrates user related and environment related factors within a unified framework to produce dynamic vendor pricing and context appropriate verbal responses. Dialogue generation in the system is supported by a large language model to simulate diverse negotiation styles for training purposes.

Index Terms: Interactive systems and tools, Virtual reality, Artificial intelligence.

1 INTRODUCTION

Negotiation is a key interactive skill in business transactions, everyday consumption, and workplace communication. However, most

existing negotiation training tools are implemented as web-based or mobile applications. Users typically complete simulated bargaining through text input fields and rule-based scripts. These tools provide limited situational information. As a result, users struggle to fully experience the dynamic changes and interaction complexity that characterize real-world negotiations.

Virtual environments offer a new opportunity to address these limitations by enabling negotiation training within structured and context-rich scenarios. We developed a Unity-based virtual negotiation training platform that places users in multi-scene virtual market environments. Within these scenarios, users engage in multiple rounds of price negotiation with virtual vendors that exhibit different bargaining styles. The system integrates user input with scene context to infer pricing preferences and negotiation tendencies. It then generates dynamic price offers and context-appropriate verbal responses. The system is designed as a training and reflection tool that allows users to repeatedly practice negotiation strategies in a controlled virtual environment, even when real-world practice opportunities are limited. Through scenario-driven multi-round interactions, the system aims to support users' understanding of negotiation processes and promote the development of self-regulation and strategic awareness.

2 METHOD

2.1 Module One: User Preference Data Acquisition

In the user preference data acquisition module, the system first records the product selected by the user and retrieves the corresponding initial list price, denoted as P_0 . Based on this information, the system processes the user input in the current round, parses the user-provided offer and related textual expressions, and extracts the current round offer, denoted as P_{user} . Combined with the dialogue context and historical interaction records, the system estimates the target price at which the user intends to close the deal, referred

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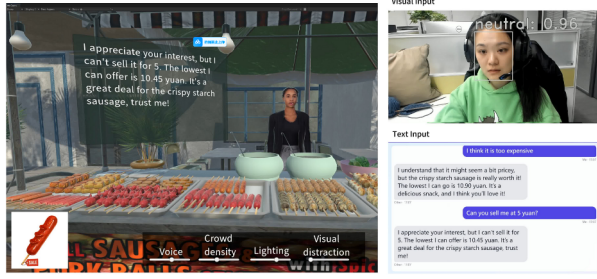


Figure 2: Overview of the proposed symmetrical reality night-market bargaining system. The platform combines first-person immersive interaction with multimodal sensing of speech, bidding history, emotional expressions, and environmental factors.

to as the ideal transaction price, as well as the minimum acceptable transaction price under the current inferred state and situational constraints.

In the visual and affective dimensions, the system incorporates facial expression-related visual cues computed using OpenCV-based image processing methods. Specifically, low-level facial features and expression-related indicators are extracted from camera input and used as auxiliary signals, rather than explicit emotion labels. These visual cues are integrated with linguistic features and historical bidding trajectories to estimate a baseline concession rate that the vendor may adopt when interacting with the user. The concession factor derived from user-specific characteristics is denoted as C_{user} , which represents the concession tendency estimated from the individual user perspective. At the same time, based on linguistic cues and visual expression indicators observed in the current negotiation round, the system selects an appropriate negotiation strategy category, such as styles emphasizing comfort, rational analysis, or a moderate sense of urgency, using predefined strategy mappings.

2.2 Module Two: Environmental Factor Detection

In the environmental factor detection module, the system models the current environmental atmosphere along multiple dimensions, including noise level, crowd density, lighting intensity, and visual distractions in the scene, while also taking into account the time pressure induced by the length of the ongoing negotiation. For example, when visual distractions are strong, it becomes more difficult for the user to carefully compare price tags and product details, and the user judgment may be weakened unconsciously, which in turn influences bargaining behavior. When the negotiation lasts for a long period, the rising time pressure allows the system to shrink the space of additional concessions according to predefined rules.

Based on these environmental signals, the system applies a set of weighted mappings that integrate the combined effects of noise, crowding, lighting, and visual distractions together with the trend of time pressure into an environment related concession factor, denoted as C_{env} . In a noisy, crowded, highly distracting environment with strong time pressure, the system may tend to increase the vendor concession rate to speed up the transaction. In a quiet, orderly environment with fewer distractions, the system may guide the vendor to adopt a more conservative concession pattern.

2.3 Module Three: Strategy Selection and Language Generation

In the strategy selection and language generation module, the system integrates the user concession factor C_{user} from the user preference module with the environment concession factor C_{env} from the environmental factor module, and computes the round concession

rate, denoted as C_{total} . Specifically,

$$C_{\text{total}} = C_{\text{user}} + C_{\text{env}}.$$

Given the initial list price P_0 , the system uses this factor to calculate the recommended selling price for the vendor in the round, denoted as P_{final} , which is used to guide the vendor pricing decision.

When generating utterances, the system feeds the negotiation strategy type, the recommended selling price P_{final} , the current user offer P_{user} , and the historical dialogue between the user and the vendor into a large language model. When the system detects that the user offer P_{user} is greater than or equal to the current acceptable deal threshold for the vendor, which is derived from P_{final} and predefined rules and denoted as P_{deal} , it determines that an agreement has been reached and terminates the current bargaining session.

When the user selects a new product, the system reinitializes the parameters associated with that product, resets time pressure and round dependent variables to their initial states, and reestimates the initial list price P_0 , the user based concession factor C_{user} , and the environment based concession factor C_{env} for the new product.

3 PLANNED EVALUATION DESIGN

We outline a comparative evaluation protocol intended to examine the potential training effects of the proposed bargaining system. In the planned study, participants would first engage in everyday real world bargaining scenarios, such as transactions in local markets, where metrics including deal success rate and relative price discount would be recorded. Participants would then complete multiple training sessions within the proposed virtual bargaining system, interacting with virtual vendors across different scenes and product types in a controlled environment.

Following the training phase, participants would return to comparable real world bargaining situations, and the same performance metrics would be collected again. The comparison between pre training and post training measures is designed to assess whether participation in the virtual training system is associated with changes in bargaining outcomes. This evaluation protocol is planned for future execution and is not reported as completed at the time of submission.

4 CONCLUSION AND FUTURE WORK

This study presents a virtual night market bargaining training system that integrates user preference modeling, environment level factor representation, and strategy selection with language generation within a unified framework. The system is designed to support interactive training and reflection on bargaining strategies in a controlled virtual setting.

Future work will focus on three directions. First, we plan to extend the system with finer grained behavioral features and temporal modeling approaches to better support the analysis of strategy adjustment over time. Second, we aim to expand beyond the night market scenario to more complex negotiation contexts such as rent, salary, and project outsourcing, in order to explore cross domain applicability. Third, we plan to conduct larger scale and longitudinal user studies with participants from diverse cultural backgrounds and varying levels of negotiation experience, following the proposed evaluation protocol described above.

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